https://doi.org/10.1038/s44386-026-00090-1

Physicochemical rule-embedded generative modeling for constrained peptide therapeutics

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Yuzhou Wang[1], Dongsheng Cao[2], Xiangzheng Fu[3] ✉, Xiangxiang Zeng[1] & Xiaojun Yao[4] ✉

Constrained peptides couple covalent topology, stereochemistry, three-dimensional structure, conformational dynamics and developability. We review physicochemical rule embedding across chemical-language, target-conditioned structural, geometric, atom-bond, ensemble and property-guided generative models, distinguishing generated molecular objects from downstream inference and predictions from prospective evidence. We identify limitations in chemical coverage, applicability, uncertainty and pipeline attribution, and routes towards identity-preserving, environment-aware and experimentally calibrated design.

Constrained peptides occupy a productive region of chemical space between small molecules and large biologics[1]. Their size and conformational plasticity can support recognition of broad, shallow, or discontinuous surfaces, including protein-protein interaction interfaces, enzyme exosites, and receptor surfaces that are often difficult for conventional small molecules to address[2]. Cyclization, stapling, disulfide formation, backbone N-methylation, D-amino acid incorporation, and related modifications can reduce conformational entropy, improve proteolytic resistance, and tune the balance between preorganization and adaptability. These features make macrocyclic and otherwise constrained peptides attractive modalities for targets that are incompletely served by either classical medicinal chemistry or large protein therapeutics[3].

This versatility also makes the design problem highly coupled. The object being proposed is not merely an amino acid sequence, but a chemically specified scaffold with covalent connectivity, stereochemistry, one or more accessible conformations, and an environment-dependent physicochemical profile[4]. Head-to-tail macrocycles, disulfide-rich peptides, stapled helices, 1,3,5-tris(bromomethyl)benzene (TBMB)-linked bicycles, side-chain lactams, and N-methylated heterochiral macrocycles impose distinct requirements on connectivity, closure geometry, local strain, side-chain packing, synthesis, permeability, and target engagement[5]. Conventional descriptors such as molecular weight, topological polar surface area, hydrogen-bond donors, lipophilicity, and rotatable-bond count remain useful, but their interpretation is scaffold- and assay-dependent[6]. Molecular chameleonicity denotes environment-dependent changes in exposed polarity and intramolecular hydrogen bonding[7]; in cyclic peptides, this behavior can help reconcile aqueous handling with membrane-compatible conformations[8]. Experimentally characterized, computationally designed macrocycles further show that conformationally programmed polar shielding can coexist with membrane permeability and oral exposure[9]. Figure 1 illustrates representative constrained-peptide architectures and their linkage chemistries.

Generative modeling has expanded the scale at which peptide-like candidates can be proposed under sequence, topology, target, or property constraints[10]. Early rule-based generators emphasized enumerability and formal validity, whereas recent approaches increasingly place chemical or physicochemical constraints within molecular notation, topology-aware structural representations, geometric or atom-bond generation, conformational sampling, and property-guided optimization. These studies broaden the accessible design space, but the claims they support depend on the chemical object represented, the stage at which a constraint acts, and the endpoint that is actually benchmarked or measured.

Generic architecture labels are therefore less informative than the molecular object represented and the point at which a rule changes the generative distribution. Foundational work on Bayesian neural networks provides methodological context for predictive uncertainty[11], while early stochastic treatments of reinforcement learning provide general context for reward-driven updating[12]; neither is treated here as constrained-peptide-specific evidence. Post hoc filtering remains valuable for triage and validation, but it is separated from constraints that directly shape sampling.

Recent reviews and perspectives have surveyed therapeutic-peptide generative models[13], computational and artificial intelligence-driven strategies for cyclic-peptide discovery[10], and generative AI in peptide-based drug design[14]. Broader work on large language models in drug discovery provides related architectural context[15]. The present scope is narrower: absorption, distribution, metabolism, excretion, and toxicity (ADMET)- and developability-related models are discussed only when they shape constrained-peptide generation, guide candidate selection, define property

[1]College of Computer Science and Electronic Engineering, Hunan University, Changsha, China. [2]Xiangya School of Pharmaceutical Sciences, Central South University, Changsha, China. [3]Faculty of Computer Science and Artificial Intelligence, Shenzhen University of Advanced Technology, Shenzhen, China. [4]Faculty of Applied Sciences, Macao Polytechnic University, Macao, China. ✉e-mail: fuxiangzheng@suat-sz.edu.cn; xjyao@mpu.edu.mo

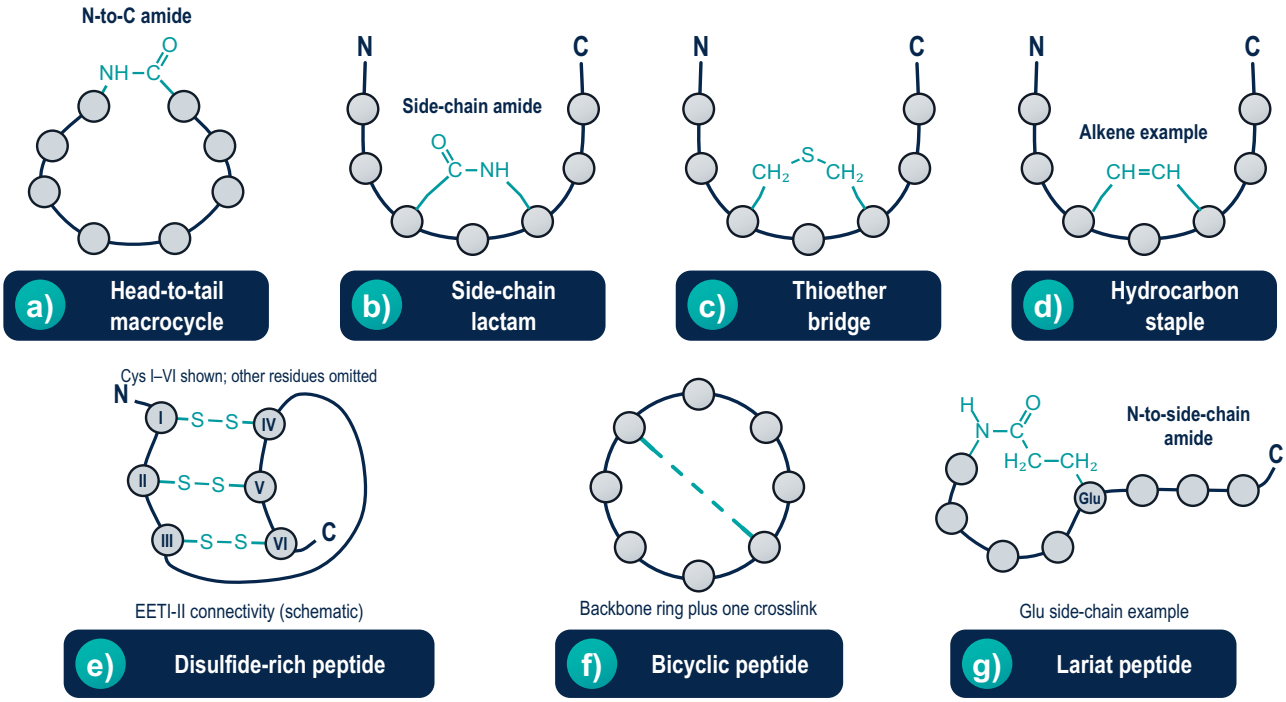


Fig. 1 | Architectures and linkage chemistries of constrained peptides. Panels show **a** a head-to-tail macrocycle with an N-to-C amide, **b** a side-chain lactam, **c** a thioether bridge, **d** a hydrocarbon staple illustrated by an alkene, **e** a disulfide-rich peptide, **f** a bicyclic peptide comprising a backbone ring and one crosslink, and **g** a lariat peptide. Gray circles denote residues; navy lines, the backbone; and teal segments, cyclization or crosslinking. The dashed link in (**f**) has unspecified chemistry. Navy N/C labels mark free termini. In (**e**), Ecballium elaterium trypsin inhibitor II (EETI-II) has disulfide connectivity I–IV, II–V and III–VI[124]; Roman numerals identify cysteines in sequence order, with intervening residues omitted. The lariat in (**g**) links the N terminus to a glutamate (Glu) side chain, leaving a C-terminal tail. Linkage atoms are shown explicitly, and side chains are abbreviated. Bond lengths, angles and backbone shapes are schematic, and categories can overlap.

preferences, or provide validation endpoints. The comparison asks how representative methods encode modified chemistry, topology, three-dimensional geometry, conformational ensembles, target context, and developability, and how strongly their outputs are substantiated. Sequence-level approaches are considered alongside residue- and atom-aware systems, while generation-time rule embedding is kept distinct from downstream scoring. The final sections identify interfaces through which noncanonical chemistry, environment-aware ensembles, calibrated multi-objective optimization, synthetic tractability, and selective experiments could be connected.

Generated objects and representation adequacy

Meaningful comparison begins with the molecular object a method actually produces. We distinguish seven levels: a residue sequence; a covalent graph that specifies monomers and inter-monomer bonds; a stereochemically resolved molecular identity; a single three-dimensional conformer; a population-weighted conformational ensemble; a set of environment-conditioned ensembles; and a target-bound complex. The levels are information-bearing rather than interchangeable. A parseable sequence does not establish linkage geometry, a valid graph does not establish low strain, a plausible conformer does not establish population weights, and an unbound ensemble does not establish a binding mode or affinity.

Representation adequacy is therefore endpoint-dependent, not a monotonic function of detail. Serialized notations can support large-scale chemical enumeration when topology and stereochemistry are unambiguous, whereas permeability, chameleonicity, intramolecular hydrogen bonding and recognition often depend on conformational populations and environment[8]. Throughout this Review, information is classified as explicit in the generated object, inferred by the model, reconstructed by a downstream physical or learned procedure, or absent. Information introduced during reconstruction is not credited to the generator, and its uncertainty must accompany subsequent property or therapeutic claims.

The same accounting defines the demonstrated chemical domain. Results for canonical head-to-tail macrocycles do not establish performance on disulfide-rich, stapled, side-chain-linked, heterochiral, N-methylated or multicyclic systems. Claims of modality-level generalization should be stratified by topology, ring size, linkage chemistry, stereochemical composition, noncanonical-residue burden and assay context. Cross-class generalization should be claimed only when performance is evaluated on prespecified held-out topology or chemistry classes and reported per class with explicit denominators, uncertainty intervals and failure rates; otherwise, conclusions should remain within the demonstrated chemical scope.

Design challenges and modelable rules

Design variables rarely change independently in a constrained peptide[16]. A residue substitution can alter charge, hydrophobicity, hydrogen-bonding capacity, secondary-structure preference, and protease susceptibility; in a macrocycle, the same edit can also change ring strain, closure geometry, side-chain packing, intramolecular hydrogen-bond networks, and access to a target-compatible pose. A string generator may preserve peptide-like composition while missing disulfide registration, lactam spacing, staple geometry, stereochemistry, or multicyclic linker compatibility[5]. Conversely, a structure generator may produce a plausible bound conformation while remaining insensitive to permeability, solubility, purification difficulty, or metabolic liability[17]. The relevant search space is therefore chemically heterogeneous, topology-dependent, and environment-sensitive, rather than reducible to a single objective.

Residue chemistry supplies the local physicochemical vocabulary for this search[16]. Charge distribution, hydrogen-bond donors and acceptors, hydrophobic and aromatic content, N-methylation, D-residue substitution, proline placement, and local amphipathicity can influence solubility, serum stability, protease resistance, cell penetration, and target recognition[18]. Orally absorbed cyclic peptides illustrate this context dependence: successful examples can exceed conventional Rule of Five limits while reducing

effective hydrogen-bond donation or exposed polarity through N-methylation, D-residues, depsipeptide bonds, proline substitutions, and intramolecular hydrogen bonding[6]. Nuclear magnetic resonance (NMR) amide temperature coefficients have also been used to identify solvent-exposed amides for selective N-methylation in an orally bioavailable cyclic-peptide series[19]. Models can encode these tendencies through residue vocabularies, motif or composition constraints, descriptor targets, validity masks, chemically defined edits, or property-conditioned objectives. These rules are best treated as scaffold- and assay-dependent biases rather than universal cutoffs[20].

Topology adds a distinct set of rules once residues are connected into a constrained scaffold[5]. Head-to-tail amides, side-chain lactams, disulfides, thioethers, hydrocarbon staples, backbone-to-side-chain bonds, and multicyclic crosslinkers are generally not recoverable from sequence adjacency alone. Modelable variables include monomer identity, bond order, linkage atoms, anchor compatibility, ring size, stereochemistry, crosslink multiplicity, and reaction feasibility. These variables can be represented through constrained vocabularies, graph annotations, linkage masks, geometry restraints, chemically valid decoding operations, or topology-conditioned sampling. The important distinction is whether topology is merely expressible in the molecular representation, selected during generation, or enforced by the sampling and decoding process. A formally cyclic string or valid graph may still correspond to strained closure geometry, incorrect stereochemistry, or an impractical synthetic route.

Cyclization alone does not resolve the conformational problem[21]. Constrained peptides often populate multiple low-energy states, and the conformation that engages a target may differ from those favored in aqueous or membrane-like environments[22]. Intramolecular hydrogen bonds, cis-trans amide isomerization, side-chain rotamers, and polar exposure can shift with solvent and binding context[23]. Relevant modelable rules therefore include closure-compatible torsions, chirality, cis-trans state, intramolecular hydrogen-bond patterns, rotamer compatibility, solvent exposure, and ensemble populations. These features can be incorporated as internal-coordinate constraints, ensemble-conditioned objectives, solvent- or state-specific descriptors, or physical refinement and validation steps. A single closed conformer can support a geometric plausibility claim, but it is limited evidence for aqueous, membrane-competent, and target-bound populations.

Developability introduces another layer of coupled and assay-specific endpoints. Permeability, solubility, proteolytic stability, metabolic liability, hemolysis, toxicity, and synthetic tractability are not independent of topology or conformation. Cyclic-peptide permeability models have been compared across scaffold and representation choices[24], while applicability-domain analysis remains essential for judging when such predictions are meaningful[20]. Related endpoints include peptide half-life, which has been modeled from blood-stability data and from enzymatic-cleavage features transferred across species and organs[25,26], and hemolytic propensity, which can be addressed with neural-network safety classifiers[27]. Models may use measured or predicted endpoints as rewards, preferences, posterior guidance, acquisition criteria, or filters. Docking energies, predicted permeability, half-life estimates, and safety scores remain surrogate outputs; improvements in these scores are computational changes rather than measured binding, permeability, stability, safety, pharmacokinetics, or therapeutic benefit, particularly outside a defined applicability domain[20] or near activity cliffs[28]. Prospective absorption, distribution, metabolism, and excretion studies show why retrospective accuracy alone is insufficient[29]. Assay identity is part of the endpoint: PAMPA and Caco-2 assays capture different aspects of transport[30], MDCK-MDR1 adds an efflux-aware cellular context[31], and bulk live-cell measurements provide another route to experimental permeability labels[32].

Rules can intervene at several points in a generative workflow. Representation determines what can be expressed; conditioning specifies target, topology, closure, property, or ensemble context; and decoding can restrict monomers, atoms, bonds, stereochemistry, or ring size. During training or sampling, denoising, flow matching, reward optimization, preference-aligned updating[33], posterior guidance, or active learning can bias the proposed distribution before candidates are finalized. Evaluation tools generally act later through scoring, filtering, relaxation, docking, retrospective benchmarking, uncertainty estimation, or experimental validation. Energy and design frameworks[34], cyclic-peptide docking benchmarks[35], peptide tertiary-structure benchmarks that include disulfide-cyclic examples[36], force-field assessments of cyclic-peptide ensembles[37], and permeability-predictor comparisons[24] illustrate how evaluation can reveal plausibility and limitations without necessarily changing the initial sampling distribution. Evidence is therefore interpreted at the level actually benchmarked or measured: computational generation, structural assessment, biochemical testing, cell-based activity, transport measurements, and prospective synthesis are treated as distinct endpoints.

A practical distinction follows. Generation-time embedding includes rules in the molecular representation, conditioning variables, decoding grammar, denoising or flow trajectory, loss, reward, preference objective, posterior guidance, or optimization path. Evaluation-time use includes scoring, filtering, relaxation, ranking, retrospective benchmarking, and experimental validation. A third role is conceptual: medicinal-chemistry or biophysical principles may motivate interpretation without being implemented in the model. Reporting these roles separately clarifies whether a rule changes the sampling distribution or is applied after candidate proposal. Table 1 summarizes the rule categories and their modelable forms.

Landscape of representative constrained peptide generative models

Figure 2 summarizes the chronological and methodological landscape, and Table 2 provides the corresponding model-level comparison. The taxonomy is organized by the molecular object principally generated and by the mechanism through which constraints act, rather than by broad labels such as Transformer, diffusion model, or flow model. Several systems span more than one capability; their placement reflects the dominant contribution rather than an exclusive classification. Throughout this Review, evidence is treated hierarchically rather than interchangeably: computational validity, retrospective benchmarking, predicted property optimization, structural validation, biochemical binding or cellular activity, permeability measurement, and prospective synthesis-and-assay validation support different levels of claim.

Five recurrent families are useful for organizing the surveyed literature. Sequence, notation, and latent-space methods operate on serialized molecular descriptions or sequence-level latent variables. Topology-aware target-conditioned systems generate or optimize bound cyclic backbones and sequence-structure solutions. Constraint-conditioned geometric and atom-bond models place closure or covalent chemistry directly in residue- or atom-level generation. Ensemble methods sample distributions of accessible conformations. Developability-guided methods alter or select candidate populations according to property preferences.

These categories are complementary rather than mutually exclusive. A notation model may also use reinforcement learning, a topology generator may include auxiliary property guidance, and a permeability-focused workflow may combine prediction, selection, and limited generation. Therefore, the placement of each method in Fig. 2 and Table 2 indicates its dominant contribution, not an exclusive capability boundary. CycloPs[38] is retained as a historical precursor because it made residue and linkage rules explicit before learned generation became common.

The detailed model-by-model discussion is reserved for the following subsections, which evaluate each family according to the generated object, the location of rule embedding, and the endpoint actually benchmarked or measured. Figure 3 compares these five core methodological families. Figure 2 additionally includes rule-based precursors and noncanonical-amino-acid structure-support methods.

Pipeline role provides a second axis. We use the term generative model for a method that proposes new instances of a declared object, such as molecular identity, covalent topology, a sequence-structure solution or a conformer, through sampling or search. An editor explores local

https://doi.org/10.1038/s44386-026-00090-1

Table 1 | Physicochemical and developability rules as modelable constraints

Rule category	Medicinal chemistry meaning	Measurable indicators	Modelable form	Possible embedding mode	Cautions
Size and flexibility	Macrocycles may exceed RO5 limits; excessive size and flexibility can reduce exposure[6]	MW, rotatable bonds, ring size, residue count	Descriptor targets or hard filters	Input conditioning, reward, Pareto constraint	Scaffold- and assay-dependent thresholds
Polarity and hydrogen bonding	Solubility and binding benefits; exposed polarity can impair passive permeability	TPSA, MPSA, HBD, HBA, exposed PSA	Conformation-aware descriptor or geometry loss	Reward, posterior guidance, ensemble ranking	Static TPSA insufficient for chameleonic macrocycles[7]
Intramolecular hydrogen bonding	NH satisfaction and polar shielding can enable membrane crossing[121]	IMHB counts, NH temperature coefficients, HDX, amide exposure	Graph/contact constraints or ensemble features	Loss, reward, sampling guidance; validation reported separately	Over-optimization may impair solubility or target binding
Lipophilicity and solubility balance	Hydrophobicity aids membrane partitioning but may impair solubility and increase toxicity[16]	cLogP, logD, ChromlogD7.4, LPE, solubility assays	Multi-objective trade-off	Pareto ranking, DPO preferences, Bayesian acquisition	Assay- and ionization-dependent interpretation
Chameleonicity	Solvent-dependent polar exposure reconciles solubility and permeability[115]	DeltaPSA, solvent-dependent IMHB, conformer populations, Rg, SASA	Multi-state conformer objective	Ensemble generation, environment-conditioned scoring	Costly measurement and simulation; force fields may bias populations
Permeability	Membrane crossing for cell entry and oral absorption; efflux and degradation liabilities	PAMPA, Caco-2, MDCK-MDR1, CAPA, Papp, CP50	Predictor, reward, acquisition target	RL, DPO, posterior guidance, presynthesis filter	Assays capture different mechanisms[20]
Proteolytic and metabolic stability	Resistance to enzymes in gut, serum, liver and cells supports exposure[26]	Half-life, cleavage propensity, serum stability, organ-specific predictors	Sequence- and conformation-aware stability score	Reward, filter, active-learning endpoint	Species-, enzyme- and organ-dependent
Safety and immunogenicity	Therapeutic liabilities: hemolysis[27], toxicity and immune recognition	Hemolysis assays, cytotoxicity, MHC binding, cytokines	Classifier or conservative filter	Reward, Pareto constraint, uncertainty-aware filter	Sparse data; costly false negatives
Synthetic feasibility	Accessible, isolable and chemically tractable candidates	Monomer availability, route length, coupling and cyclization yield, purity	Peptide-specific synthetic grammar	Constrained vocabulary, route-aware reward, experimental feedback	Small-molecule SA scores such as SCScore[118] miss SPPS and purification constraints

RO5 rule of five, *MW* molecular weight, *PSA* polar surface area, *TPSA* topological PSA, *MPSA* molecular (three-dimensional) PSA, *HBD/HBA* hydrogen-bond donor/acceptor, *IMHB* intramolecular hydrogen bond, *HDX* hydrogen–deuterium exchange, *LPE* lipophilic permeability efficiency, *DeltaPSA* solvent-dependent change in PSA, *Rg* radius of gyration, *SASA* solvent-accessible surface area, *PAMPA* parallel artificial membrane permeability assay, *MDCK-MDR1* Madin–Darby canine kidney cells expressing multidrug resistance protein 1, *CAPA* chloroalkane penetration assay, *Papp* apparent permeability coefficient, *CP50* concentration giving 50% HaloTag occupancy in CAPA, *ChromlogD7.4* chromatographic logD at pH 7.4, *MHC* major histocompatibility complex, *SA* synthetic accessibility, *SCScore* synthetic complexity score, *SPPS* solid-phase peptide synthesis.

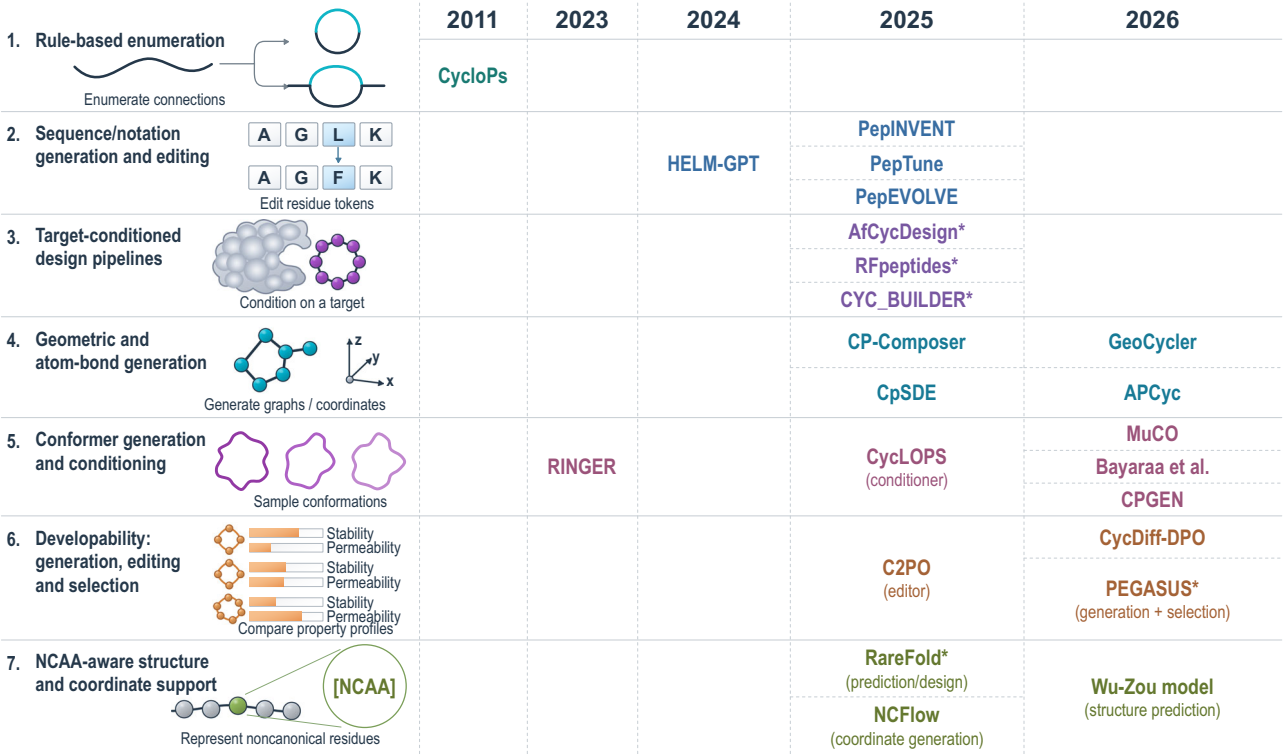


Fig. 2 | Chronological landscape of constrained-peptide design methods. Methods are grouped by their dominant contribution; categories can overlap, and parenthetical labels specify supporting roles. Years follow the cited journal issue or, where unassigned, the cited preprint. The year columns are discrete and not time-scaled. Asterisks indicate selected, endpoint-specific experimental evidence, not validation of every capability or candidate. Miniatures illustrate category-level concepts rather than individual implementations or measured performance. NCAA, noncanonical amino acid. Colors distinguish method categories. A, G, L, K and F denote alanine, glycine, leucine, lysine and phenylalanine, respectively.

transformations around an existing candidate; a predictor maps a fixed candidate to an endpoint; a structure predictor or refiner assigns coordinates without necessarily changing molecular identity; and a selector ranks an already proposed set. When these components alter sampling or search through rewards, preferences or guidance, the overall workflow is a generative hybrid, but its constituent contributions remain separable. For coordinate-sampling methods with a fixed molecular identity, classification follows the declared target: sampling a conformational distribution is conformer or ensemble generation, whereas using stochastic samples to infer or refine a representative task-specific structure remains structure prediction or refinement. Role assignments therefore follow the object and endpoint being explored, not the presence of a diffusion process, neural network or optimization loop.

Sequence, notation, and latent-space generation

Sequence and notation models gain scale by treating a constrained peptide as a writable chemical language. HELM-GPT[39] places rules in a bounded monomer vocabulary and explicit bond syntax, whereas PepINVENT[40] uses a peptide-oriented chemical grammar derived from monomer simplified molecular-input line-entry system (SMILES) strings. SELFIES provides a useful small-molecule precedent for syntax-robust molecular strings[41], but robust serialization alone does not supply peptide topology, monomer provenance, or three-dimensional closure. This compression supports large-library generation and medicinal-chemistry editing, while ring strain, target geometry, and environment-dependent conformations remain implicit unless connected to a structural model.

HELM-GPT[39] occupies the bounded-vocabulary end of this design space. HELM keeps monomer identity and inter-monomer bonds explicit, so predefined noncanonical building blocks and macrocyclization remain visible to the autoregressive model; REINVENT-style reinforcement learning[42] then shifts the generated distribution using permeability and KRAS-affinity predictors. The monomer library acts as a practical chemical prior, but it also limits exploration to the available vocabulary. Within the demonstrated workflow, reported property gains are predictor outputs, and native three-dimensional structure generation or measured affinity and permeability were not reported. The method is therefore most useful as an auditable chemical-language layer.

PepINVENT[40] exposes more of each monomer's chemical structure through CHUCKLES. Masked infilling can edit canonical residues and noncanonical amino acids, including D-residues, N-methylation, and cyclic connectivity, while retaining a peptide-oriented grammar. Relative to a fixed monomer token, the SMILES-derived description permits finer chemical edits but increases the burden of valence, stereochemical, and synthetic checks. Reinforcement-learning scores can guide the policy; based on the reported evidence, the principal advance is representational flexibility rather than solved geometry or measured developability.

The comparison is therefore not simply HELM versus CHUCKLES, but curated monomer semantics versus more exposed submonomer chemistry. HELM-GPT makes the available building blocks and connectivity readily auditable, whereas PepINVENT gives the generator greater freedom to modify molecular structure. Neither notation alone is sufficient to establish stereochemical fidelity after decoding, feasible three-dimensional closure, a low-energy ensemble, or a target-compatible pose. These boundaries reflect the generated object rather than a failure of chemical notation.

Latent-space methods inherit similar boundaries unless the latent variables are trained against topology, chemistry, and geometry that matter for constrained peptides. Smooth interpolation or sequence reconstruction does not ensure that distance in latent space corresponds to closure feasibility, conformational similarity, or synthetic accessibility[43]. PepVAE demonstrated latent navigation and activity prediction for linear antimicrobial peptides[44]. A broad-spectrum antimicrobial-peptide foundation

https://doi.org/10.1038/s44386-026-00090-1

Table 2 | Representative generative, optimization and design-support models relevant to constrained-peptide therapeutics, classified by primary output object, chemical or design scope and pipeline role

Model	Primary output object and scope	Explicit representation/ reconstruction	Pipeline role/mechanism	Rule embedding/selection	Evidence/applicability boundary
CycloPs[38] (2011)	Cyclized or constrained virtual libraries; user-defined residues and linkages, including NCAAs	Residue and linkage rules; externally reconstructed 3D	Generator: combinatorial enumeration	User-specified residue/linkage constraints	Computational libraries; no learned structure, target or developability model
HELM-GPT[39] (2024)	Macrocycle strings with NCAAs; de novo generation and property optimization	HELM monomers and bonds; SMILES derived downstream	Generator/optimizer: decoder-only GPT with RL	Constrained HELM vocabulary; predictor rewards	Computational validity/property scores; no 3D geometry or assays
PepINVENT[40] (2025)	Linear or cyclic peptide strings with NCAAs; masked completion and optimization	CHUCKLES monomers; atom graph reconstructed for scoring	Generator/optimizer: Transformer infilling with REINVENT RL	CHUCKLES grammar; ring, solubility and permeability scores	Computational scores; no conformation; novel NCAAs need synthesis checks
PepTune[92] (2025)	Therapeutic peptide SMILES; multi-objective design with cyclic modifications and NCAAs	Peptide SMILES; RDKit ETKDG conformer reconstruction; AutoDock Vina pose scoring	Generator/optimizer: masked discrete diffusion with MCTG	Bond-dependent masking; invalid loss; Pareto rewards	Predictor and docking benchmarks; no assays; predictor and topology limits
PepEVOLVE[91] (2025)	Position-edited peptide strings; macrocyclic/linear lead optimization	CHUCKLES strings; externally predicted properties	Editor/optimizer: Transformer, bandit routing and RL	Position-aware masking; multi-objective reward	PepINVENT comparison; no 3D or assays; collapse in some settings
AfCycDesign[56] (2025)	Designed sequences with AF2-predicted structures and binding modes; head-to-tail 7–16-mers	Sequence and cyclic offset; AF2-inferred coordinates	Hybrid: AF2 design or hallucination plus structure prediction	Cyclic offset in relative positional encoding	X-ray structures and binding assays; canonical head-to-tail scope; no permeability
RFpeptides[57] (2025)	Generated backbones and designed sequences; head-to-tail macrocyclic binders	Backbone coordinates and cyclic RPE; ProteinMPNN-derived sequences	Generative pipeline: RFdiffusion backbones plus sequence design	Cyclic RPE; target/hotspot conditioning	SPR, competition and X-ray evidence; no noncanonical chemistry or developability; post hoc affinity filtering
CYC_BUILDER[60] (2025)	Assembled peptide-target complexes; canonical head-to-tail or disulfide-cyclized binders	Interface fragments and coordinates; closure reconstructed during assembly	Generator: MCTS fragment assembly	Binding, stability and cyclization rewards	TNF-alpha binding and cellular functional assays; canonical residues; low-micromolar hits
CP-Composer[73] (2025)	Target-conditioned sequence-structure graphs; stapled, head-to-tail, disulfide and bicyclic peptides	Residue graph and coordinates; atomistic chemistry reconstructed downstream	Generator: geometric latent diffusion	Type and distance constraints; CFG	Closure metrics, selected MD; canonical residues; no assays
GeoCycler[74] (2026)	Constraint-conditioned residue-level cyclic structures; several prescribed topologies	Residue-level latent geometry; atomistic chemistry reconstructed downstream	Generator: reward-aligned diffusion	Type-gated stair reward; CFG	Closure and pass@k ablations; bicyclic cases more challenging; no experiments
APCyc[75] (2026)	Topology-conditioned sequences and residue-level 3D structures; amide, disulfide and isopeptide cyclization	Residue tokens and linkage matrix; all-atom chemistry reconstructed downstream	Generator: latent diffusion with topology heads	Topology prediction, validity mask and Bayesian property guidance	Large computational benchmark; limited chemistries; no synthesis; predictor-based properties
CpSDE[76] (2025)	Atom-bond sequence-structure complexes; multiple cyclization types	Atom-bond graph and coordinates; residue sequence routed	Generator: harmonic SDE with residue router	Graph-Laplacian harmonic prior; chemical graph	Computational benchmark and MD cases; canonical router; no assays
MuCO[79] (2026)	All-atom cyclic conformers; head-to-tail, disulfide and isopeptide peptides	Backbone frames and torsions; side chains generated then refined	Ensemble generator: SE(3) and torsional flow matching	Cyclic RPE; CPSea training; force-field refinement	Energy and diversity metrics; noncanonical, target-binding and developability scope not demonstrated
RINGER[77] (2023)	Conformer ensembles of homodetic 4–6-mer macrocycles	Internal coordinates; Cartesian reconstruction and closure post hoc	Ensemble generator: diffusion Transformer	Internal-coordinate bias; cyclic graph distances	Metadynamics and simulation-derived ensemble benchmark; post hoc closure; limited chemistry
Bayaraa et al.[83] (2026)	Backbone ensembles; mixed-chirality cyclic 5–7-mers	Backbone dihedrals and chirality; no side chains or atomistic closure	Ensemble generator: DDPM with U-Net	MD-trained dihedral distributions	MD ensemble comparison; no side chains, explicit closure, target conditioning or property objectives
CycLOPS[82] (2025)	Crosslinked-macrocycle ensembles conditioned from linear peptides	All-atom ensembles; conditioned from a pretrained linear-peptide flow	Ensemble conditioner: conditioned normalizing flows	Cyclization loss from tetrahedral geometry	Computational validity and loss analysis; validity and steric limitations; no explicit NCAA generation

Table 2 (continued) | Representative generative, optimization and design-support models relevant to constrained-peptide therapeutics, classified by primary output object, chemical or design scope and pipeline role

Model	Primary output object and scope	Explicit representation/ reconstruction	Pipeline role/mechanism	Rule embedding/selection	Evidence/applicability boundary
CycDiff-DPO[95] (2026)	Target-conditioned sequence-structure candidates; head-to-tail binders optimized for predicted permeability	Residue graph and coordinates; surrogate-predicted properties	Generator/optimizer: latent diffusion with DPO	Predictor-derived permeability preferences	Computational benchmark and MD; predicted permeability only; canonical head-to-tail scope
C2PO[96] (2025)	Edited cyclic-peptide atom graphs; permeability optimization of existing candidates	Atom graph editing and repair; stereochemistry not retained; no 3D	Editor/optimizer: graph estimator, HotFlip and beam search	In-loop permeability estimator; post hoc chemical-validity repair	Computational only; predicted gains depend on estimator and repair; no prospective permeability assay
PEGASUS[31] (2026)	VAE-sampled cyclic candidates; assay-informed permeability optimization of polar peptides	Sequence, SMILES, graphs, QM features; predictor-derived endpoints	Predictive/generative hybrid: VAE exploration plus predictor prescreening	Physicochemical and chameleonicity criteria mainly in predictors, filters and selection	Prospective synthesis, solubility and MDCK-MDR1 permeability assays on selected designs; limited target, topology-general or ensemble claims
CPGEN[78] (2026)	Heterochiral cyclic ensembles and designed sequences; ensemble-based inverse design	C-alpha coordinates, latents, sequence; reconstructed atomistic ensembles	Ensemble generator/inverse design: partially latent flow matching	Chirality vocabulary; multimodal CFG	Ensemble metrics versus Rosetta-sampled references; speed versus Rosetta GenKIC; no linkers, N-methylation, property optimization or experiments
RareFold[66]/ NCFlow[67] (2025)	RareFold: NCAA-containing structure prediction; NCFlow: atom-conditioned coordinates or complexes; NCAA-aware design support	RareFold: NCAA tokens and predicted coordinates; NCFlow: atom graph and generated coordinates	RareFold: prediction and design support; NCFlow: atom-conditioned generation	NCAA vocabulary; atom-bond and pocket conditioning	RareFold/EvoBindRare: SPR for selected NCAA binders; NCFlow: computational only; topology and stereochemical coverage method-dependent

The Primary output object and scope column separates the object returned by each method from its declared chemical or design scope. Pipeline roles are defined by the object that changes. A generator creates a new molecular identity, topology, coordinate set or distribution; an editor or optimizer modifies an existing object under an objective; a supporting method predicts, folds, refines or ranks without creating that object; and a hybrid combines these roles. The Evidence/applicability boundary column reports both validation level and principal scope limit. Computational validity, retrospective benchmarking, predicted property optimization, structural validation, binding or cellular assays, measured permeability, and prospective synthesis-and-assay validation support different claims and are not interchangeable.

model supplies transferable sequence representations[45], SQ-DiffuPep adds multimodal latent diffusion[46], and Diff-AMP integrates generation, identification, prediction, and optimization[47]. These studies provide useful algorithmic background, but their mainly linear, canonical peptide domains do not by themselves demonstrate constrained topology, noncanonical chemistry, macrocyclization geometry, or chameleonic ensemble design.

For sequence-level models, the claim should match the test. A valid grammar shows that a string can be parsed under the chosen representation. A retrospective predictor benchmark describes performance on a defined data split. A higher predicted reward shows that the generator found regions favored by that predictor. These outcomes provide limited evidence for measured binding, permeability, or therapeutic improvement when the generated chemistry extends beyond the predictor's applicability domain[20] or when benchmark performance varies sharply across data splits and representations[24].

These approaches may be most useful as the chemical-language layer of a broader design stack. They provide scalable residue planning, explicit monomer identity, and a natural interface for medicinal-chemistry edits. Their translational value would increase if symbol-level decisions were propagated without loss into atom-level connectivity, stereochemistry, closure-compatible geometry, conformational ensembles, and assay-aware selection.

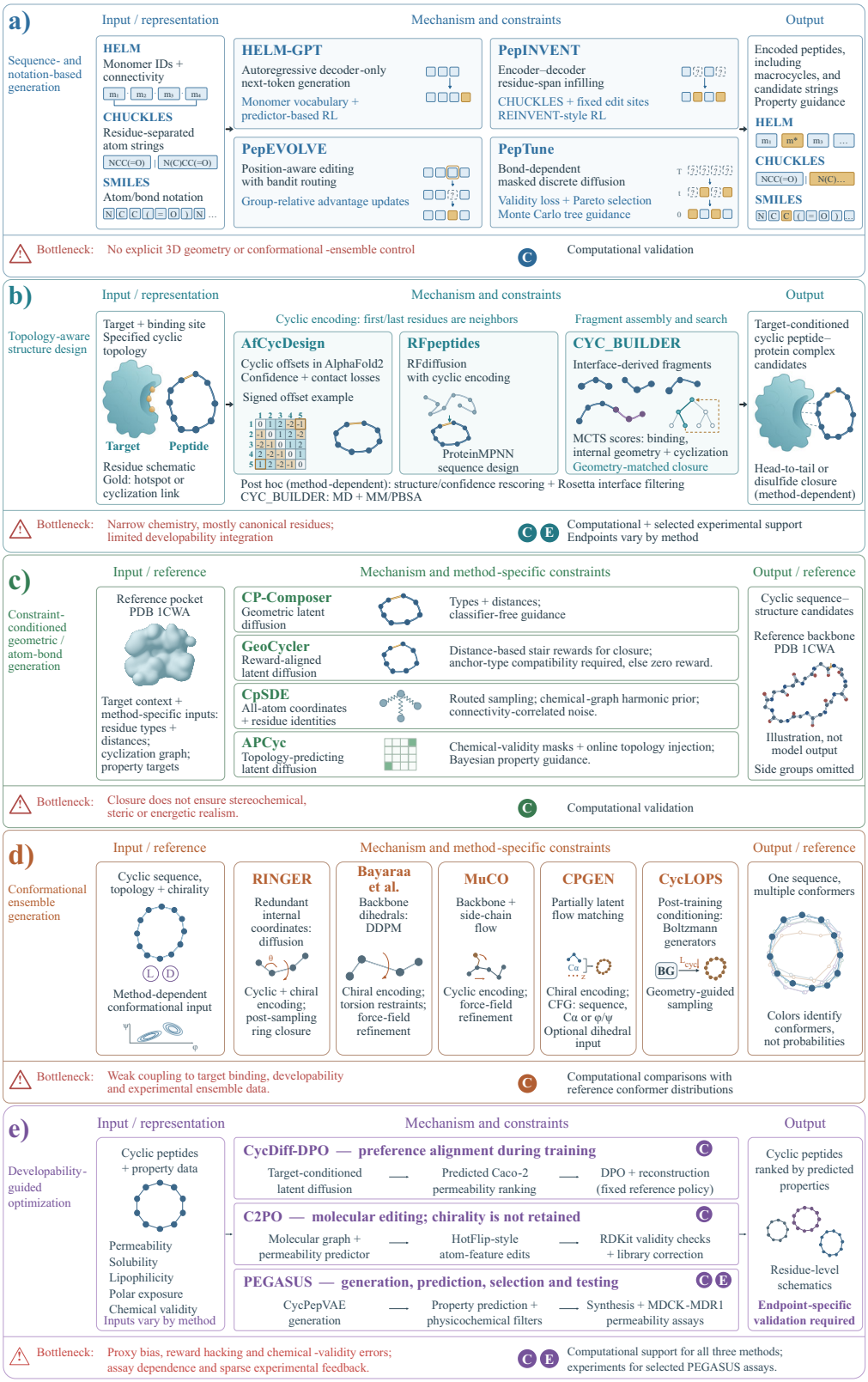
Topology-aware target-conditioned structure design

Topology-aware target-conditioned design offers a comparatively direct route from a generated object to an experimentally testable binding hypothesis. Earlier structure-based pipelines established complementary precedents: exhaustive design of ordered macrocycles[48], structure-guided anchor extension into enzyme active sites[49], rapid template matching with cPEPmatch[50], and fragment-based assembly with Des3PI[51]. Generic target-aware peptide generators such as PepFlow[52], PPFlow[53], PepGLAD[54], and DiffPepBuilder[55] contribute transferable geometric priors, but cyclization is not uniformly native to their representations. In the models emphasized here, cyclization enters the structure representation or the search over bound geometry rather than being checked only after a linear binder has been proposed. Their binding evidence is interpreted within that context and is not generalized to permeability, stability, or broader pharmacology.

AfCycDesign[56] modifies AlphaFold2 relative positional encoding with a cyclic offset matrix so that the N- and C-terminal residues are treated as neighbors during structure prediction and sequence optimization. The generative contribution lies in fixed-backbone design and hallucination under this topology-aware representation; the structure-prediction component is supporting technology rather than generation by itself. Experimentally solved structures of designed macrocycles support fold accuracy, while binding assays on target-directed designs support target engagement. Within the demonstrated scope, the method provides a clear route to canonical head-to-tail cyclic structures, although ensemble and developability optimization are outside the reported workflow.

RFpeptides[57] carries cyclic positional encoding into an RFdiffusion-based binder-design pipeline. RFdiffusion supplies the protein-structure denoising prior[58], while ProteinMPNN performs sequence design on generated backbones[59]. Target structures or hotspot residues condition macrocyclic backbone generation, followed by computational filtering. Surface plasmon resonance and competition assays establish biochemical binding for selected designs, and several X-ray complex structures directly test the proposed bound geometries. This evidence is strongest for target-conditioned structural competence within canonical head-to-tail macrocycles, a dominant bound state, and post-generation interface selection.

CYC_BUILDER[60] reaches a related endpoint through fragment assembly and Monte Carlo tree search rather than global denoising. Binding, intramolecular stability, and cyclization propensity contribute to the search reward, so these criteria affect growth rather than appearing only as terminal filters. The framework supports head-to-tail and disulfide closure in a canonical residue space. For selected TNF-α designs, biochemical binding and cellular reporter or transcriptional responses



provide prospective functional evidence, whereas post-generation molecular dynamics and energy calculations remain computational support rather than structural validation of the designed complex. HighPlay[61] provides an adjacent reinforcement-learning example for disulfide-constrained sequence design with a narrower reported chemical and evidentiary scope.

Together, these systems show how protein-design priors can make cyclic target engagement experimentally testable. Cyclic positional encoding in AfCycDesign[56] and RFpeptides[57] is a compact inductive bias for head-to-tail closure, while fragment assembly in CYC_BUILDER can impose local closure geometry during growth. CycleDesigner[62] adapts RFdiffusion and cyclic structure prediction in a computational binder-design workflow, and

Fig. 3 | Mechanistic taxonomy of constrained-peptide modeling. Panels compare **a** notation-based generation, **b** topology-aware target-conditioned design, **c** constraint-conditioned geometric and atom-bond generation, **d** conformational ensemble generation and **e** developability-guided optimization. Routes are alternatives, not one shared pipeline. C/E badges indicate computational/selected experimental support; evidence is endpoint-specific and predictions are not measurements. C2PO edits candidates; CycLOPS conditions generators; PEGASUS couples generation, prediction, selection and assays. In (**a**), blue, amber and dashed tokens denote retained, new/edited and masked content; m labels monomer identifiers, question marks masks, and T/t/0 diffusion steps. In (**b**), gold patches mark hotspots; matrix entries show signed cyclic offsets. The backbone and receptor-pocket illustrations in (**c**) use cyclophilin A–cyclosporin A coordinates (Protein Data Bank, PDB 1CWA)[125]; they are structural references, not generated outputs or validation results. Backbone side groups are omitted. In (**c**), gray, blue and red atoms denote carbon, nitrogen and oxygen; gold links indicate cyclization. Bead chains schematize residues. Conformers in (**d**) share one sequence; their colors encode neither elements nor populations/energies. D/L denote residue configurations, Cα alpha carbon, φ/ψ backbone dihedrals, θ a local angle, z latent variables, BG Boltzmann generator, L_{cyc} cyclization guidance loss, HELM hierarchical editing language for macromolecules, CHUCKLES residue-separated chemical notation, SMILES simplified molecular-input line-entry system, RL reinforcement learning, MCTS Monte Carlo tree search, MD molecular dynamics, MM/PBSA molecular mechanics/Poisson–Boltzmann surface area, DDPM denoising diffusion probabilistic model, CFG classifier-free guidance, DPO direct preference optimization, VAE variational autoencoder, MDCK-MDR1 Madin–Darby canine kidney cells expressing MDR1/P-glycoprotein. Only selected PEGASUS peptides underwent solubility and MDCK-MDR1 permeability testing. Related property-guided methods include HELM-GPT, PepINVENT, PepEVOLVE, PepTune and APCyc.

CyclicMPNN[63] fine-tunes inverse folding for head-to-tail cyclic backbones. As currently implemented, neither mechanism class provides a general chemical-graph representation. Side-chain lactams, thioethers, hydrocarbon staples, multicyclic linkers, backbone N-methylation, D-residues, and broader peptidomimetic chemistry generally require additional representation and stereochemical control.

The evidence also remains modality-specific. A solved design structure supports fold accuracy, as demonstrated for AfCycDesign; a solved target complex additionally supports the proposed binding mode, as in RFpeptides; and a cellular response adds functional context, as in CYC_BUILDER, but may reflect uptake, stability, target engagement, and downstream biology simultaneously. Within the surveyed studies, these target-conditioned systems provide some of the clearest structural and binding validation, while solution ensembles, permeability, stability, and pharmacokinetics require separate evidence.

Noncanonical design-support methods can broaden the chemical envelope without automatically becoming constrained-peptide generators. RoseTTAFold All-Atom[64] and AlphaFold3[65] provide general all-atom biomolecular priors, but cyclic topology and modified residue chemistry still require dedicated adaptation. RareFold[66] extends residue-token prediction and design to selected noncanonical amino acids. NCFlow[67] predicts coordinates for chemically specified noncanonical residues, whereas UNAAGI[68] generates side-chain identities and coordinates at atom level. NCPepFold[69], HighFold3[70], CyclicBoltz1[71], and the stereogenic all-atom diffusion model of Wu and Zou[72] address cyclic-peptide structure prediction with increasingly explicit chemistry. When cyclization is external to the representation or the generated scaffold is fixed, these tools are better described as prediction, folding, or chemical-support technologies. These capabilities broaden design support but do not by themselves establish de novo generation of complete constrained-peptide scaffolds.

This boundary exposes an interface problem between protein-design priors and medicinal chemistry. A target-conditioned model may propose a persuasive canonical binder without choosing among alternative closure chemistries, preserving noncanonical stereochemistry, or reasoning about solvent-dependent polar exposure. Connecting target geometry to explicit covalent and ensemble models is therefore a key requirement for expanding beyond protein-like macrocycles.

Within their demonstrated scope, AfCycDesign, RFpeptides, and CYC_BUILDER support target-conditioned structural or functional competence in experimentally tested cyclic peptides. Based on the surveyed evidence, broader claims across constrained-peptide chemistry, multi-state behavior, or developability would require additional representation and assay support. Each model's contribution is best interpreted at the level of the representation and experiment actually used.

Constraint-conditioned geometric and atom-bond generation

A second line of work makes closure or covalent chemistry the principal control variable. CP-Composer[73] uses composable type and distance constraints, GeoCycler[74] aligns diffusion with topology-specific rewards, APCyc[75] predicts cyclization variables jointly with sequence and structure, and CpSDE[76] incorporates atom-bond connectivity through a harmonic stochastic differential equation. Their shared contribution is generation-time control of closure and connectivity; within the reported literature, validation is predominantly computational.

CP-Composer decomposes stapled, head-to-tail, disulfide, and bicyclic designs into composable residue-type and pairwise-distance constraints within a geometric latent diffusion model. The underlying geometric design setting is related to PepGLAD[54], while DiffPepBuilder[55] serves as a specialized comparison for disulfide-constrained design. The constraints alter the denoising trajectory and can be combined at inference, enabling zero-shot application beyond the linear-peptide training distribution. Closure success and selected molecular dynamics analyses support geometric plausibility, while the outputs remain canonical and no synthesis, binding, or permeability assays are reported. Within this scope, the method provides a clear demonstration of composable geometric control.

GeoCycler shifts the same class of closure signal from inference-time guidance toward training-time reward alignment. Its type-gated reward activates distance shaping only when compatible anchor identities are present, reducing misleading gradients from chemically incompatible pairs. Improvements in pass@k closure metrics support stronger alignment of the learned distribution to prescribed topologies. Based on the reported evidence, difficult multicyclic cases remain more challenging, and synthesis, binding, and developability were not evaluated experimentally; the main advance is reward-aligned topology control.

APCyc treats cyclization type and linkage sites as variables predicted jointly with sequence and three-dimensional structure. Extended residue tokens, a linkage matrix, compatibility masks, and equivariant updates make topology an explicit part of denoising, while Bayesian posterior guidance can bias selected property surrogates. The central contribution is automated topology-sequence-structure generation. Property-guided results are best interpreted as an auxiliary computational demonstration because the objectives are proxies, experimental calibration of uncertainty was not reported, and no prospective synthesis or assay was included.

CpSDE moves further toward chemical explicitness by representing cyclization with an atom-bond graph and incorporating that graph into a harmonic stochastic differential equation through a Laplacian-derived prior. Routed sampling alternates all-atom structural denoising with residue prediction and supports several backbone- and side-chain-linked closure patterns in one framework. Computational energy, affinity, and molecular dynamics analyses suggest useful structural behavior, while the residue router remains canonical and the demonstrated treatment of stereochemistry and clashes is limited. CpSDE is therefore best viewed as an atom-bond generative advance whose broader noncanonical and experimental scope remains to be demonstrated.

Conformational ensemble generation

A closed structure is not equivalent to a conformational model. Molecular dynamics studies show that cyclic peptides can populate multiple solution states[21], while experimental and computational analyses link solvent-

dependent conformations to chameleonic behavior[23] and membrane transit[22]. Ensemble methods therefore target a distribution of accessible states rather than a single bound or minimum-energy conformation. Their quality depends on population fidelity and physical plausibility, not diversity alone.

RINGER[77] generates macrocyclic conformers from two-dimensional inputs by diffusing redundant internal coordinates with cyclic positional encoding. Internal coordinates provide a strong inductive bias for torsional coupling and chirality, and reported benchmarks show good agreement with simulation-derived conformer ensembles for small homodetic macrocycles. Ring closure is nevertheless completed through post-generation optimization, and the demonstrated chemical domain is limited. Within that domain, RINGER is best positioned as an efficient ensemble generator with partial, rather than fully native, closure enforcement.

CPGEN[78] uses partially latent flow matching to generate heterochiral cyclic-peptide ensembles and to infer sequences from Cα geometry or backbone-dihedral distributions. Sequence, coordinate, and Ramachandran-distribution conditioning enter through multimodal classifier-free guidance, and D-residues are represented explicitly. The method's distinctive contribution is ensemble-conditioned sequence design, not property optimization. Its current evidence is computational and based on Rosetta-derived reference ensembles; linkers, N-methylation, environment-specific populations, and experimental observables remain outside the demonstrated scope.

MuCO[79] decomposes conformer generation into topology-aware backbone sampling, generative side-chain packing, and force-field refinement. Training on cyclic structures from CPSea supplies a topology-rich structural prior[80], while the final physics-based stage uses a protein force field related to CHARMM36m[81] to improve local packing and energetic plausibility. This modular design illustrates how learned and physical components can share the task, although the force field remains an external refinement layer. Based on the reported implementation, the method is best read as a computational conformer-generation workflow rather than as target-specific binder generation or permeability optimization.

Other approaches provide complementary ensemble primitives. CycLOPS[82] introduces cyclization losses into Boltzmann generators without retraining, although sample validity and steric quality can remain limiting. The Bayaraa diffusion model[83] reproduces mixed-chirality backbone-dihedral distributions efficiently, while side-chain treatment and explicit closure remain limited. CyclicCAE[84] learns a conformational latent space with terminal bond-geometry constraints. MacroConf[85] provides a dataset and workflow for evaluating macrocycle conformer generators against solution NMR data. These complementary approaches and resources support the feasibility of learning or steering ensemble distributions, while end-to-end constrained-peptide therapeutic design requires additional coupling to chemistry, target, and assay evidence.

For ensemble generation, comparison with simulation is a retrospective computational benchmark rather than experimental validation. CREMP provides a large conformer-rotamer resource for macrocyclic-peptide machine learning[86], but reference ensembles inherit force-field and sampling assumptions. Direct force-field assessment shows that population accuracy can vary substantially across cyclic peptides[37]. Conformational sampling studies also show that selecting relevant macrocycle states is environment- and flexibility-dependent[87], and permeability ranking can require ensemble-derived three-dimensional descriptors when two-dimensional descriptors cannot distinguish stereoisomers[88]. More persuasive evaluation would connect generated populations to independent NMR, crystallographic, hydrogen-exchange, or permeability-related observables; enhanced-sampling studies provide one computational bridge but not experimental proof[89].

Developability-guided optimization

When generation is coupled to property scores, the central question becomes whether the estimator is reliable in the chemical and assay domain being optimized. Cyclic-peptide permeability models can fail outside a defined applicability domain[20], and their relative rankings depend on data split, representation, and assay composition[24]. Optimization can also exploit activity cliffs[28] or retrospective artifacts that do not survive prospective ADME testing[29]. Accordingly, an increase in a predicted objective is evidence of surrogate optimization, not experimentally validated therapeutic improvement, unless the corresponding endpoint is measured prospectively.

Property guidance is mediated by a scoring engine whose representation, labels and interface can be as consequential as the generator. Predictors may enter generation as black-box rewards, differentiable guidance, preference labels or acquisition functions, whereas terminal filters act downstream on already proposed candidates. Repeated queries during reinforcement learning, tree search or preference optimization exert stronger selection pressure than one-off filtering and can amplify calibration error or activity cliffs[90]. Training labels, split strategy, assay provenance, applicability domain, uncertainty estimate and query budget should therefore be reported for any predictor used in the design pipeline, together with the point at which it intervenes.

Available scoring engines encode different approximations to peptide chemistry. Descriptors and fingerprints are inexpensive and often effective within a restricted scaffold series, but they compress conformational heterogeneity into static features. Sequence and SMILES encoders learn distributed context, graph networks retain explicit connectivity, and three-dimensional or ensemble-derived models can capture geometry-dependent exposure at greater data and computational cost[24]. Hybrid systems combine these representations with assay metadata or physical descriptors[31]. Architecture should be matched to the endpoint and chemical domain rather than treated as an interchangeable reward layer.

PepEVOLVE[91] and PepTune[92] illustrate two sequence-level routes to property-guided search. PepEVOLVE directs edits through position-aware routing and relative-advantage updates, whereas PepTune combines peptide-aware discrete diffusion with tree-guided maintenance of a predicted Pareto set. In both cases, multiple objectives alter generation rather than merely rank a fixed library. Their evidence remains computational and predictor-dependent; within the reported studies, constrained-peptide-specific three-dimensional topology, closure geometry, and ensemble competence are not established.

For PepTune, this interface is explicit. Its MCTG scoring suite uses endpoint-specific models whose outputs guide tree search and shape the predicted Pareto set. Solubility, hemolysis and non-fouling are scored by XGBoost classifiers trained on PeptideCLM peptide-SMILES embeddings; membrane permeability is scored by an XGBoost regressor using PeptideCLM embeddings; and binding affinity is estimated by a cross-attention regressor that combines PeptideCLM peptide embeddings with ESM-2 target-protein embeddings. PeptideCLM provides a peptide-aware chemical representation for modified and cyclic structures[93]. PeptideCLM-2 extends the family through scaled BERT-style SMILES encoders and pretraining objectives that combine masked-token learning with physicochemical-descriptor regression[94]. The later model broadens endpoint coverage but was not part of the reported PepTune pipeline. Neither encoder explicitly resolves three-dimensional or environment-dependent conformational ensembles, so improved endpoint prediction should not be read as evidence of a conformational mechanism.

CycDiff-DPO[95] applies preference alignment to a target-conditioned geometric macrocycle generator using rankings from a Caco-2 permeability predictor. The optimization follows the direct-preference principle introduced for language models[33], while reconstruction regularization helps preserve sequence-structure fidelity and computational binding metrics. The reported gains concern predicted permeability and related docking, energy, and molecular dynamics analyses; permeability was not measured experimentally. Relative preference optimization may reduce sensitivity to absolute score scale, but it does not by itself correct calibration error, scaffold bias, applicability-domain limitations, or activity cliffs in the underlying predictor.

APCyc[75] is a boundary case between topology generation and developability guidance. Its posterior guidance allows selected property

surrogates to influence sampling, but the model's main advance is joint prediction of cyclization topology, sequence, and structure. The property results are best read as a computational test of auxiliary guidance rather than as evidence for a validated developability platform.

C2PO[96] is better characterized as an estimator-driven editor or property optimizer than as a de novo constrained-peptide generator. It starts from a cyclic peptide and uses HotFlip-style gradients[97] from a permeability estimator to propose atom-level chemical modifications. Gradient-guided edits may improve predicted permeability but can produce invalid structures. The workflow applies chemical-validity checks and library-based correction; however, its outputs do not retain stereochemical information. C2PO is informative as a case study in predictor-guided editing rather than evidence of measured permeability improvement or topology-aware generation.

PEGASUS[31] is best treated as an assay-rich, permeability-focused predictive/generative hybrid and developability case study. It combines large-scale proxy-assay data and Caco-2-labeled sets with graph and quantum-mechanical features and a sequence variational autoencoder (VAE) for candidate exploration and prescreening. Its main value lies in assay-informed permeability modeling, candidate exploration, and measured permeability evidence; physicochemical and chameleonicity criteria act mainly through prediction, filtering, and selection rather than strongly rule-embedded generation. Prospective synthesis, solubility measurements, and MDCK-MDR1 permeability assays for selected designs provide stronger evidence than predicted scores alone. The study does not, however, establish a general solution to topology-aware generation, target-specific binding, or conformational-ensemble design.

These examples are revisited at the cross-model level below. Within the present family, the key distinction is whether a developability signal merely ranks a proposed library or actively changes editing, denoising, preference alignment, posterior guidance, or candidate exploration. HELM-GPT[39] and PepINVENT[40] illustrate predicted property enrichment without measured developability, CycDiff-DPO[95] and C2PO[96] illustrate estimator-driven optimization without prospective permeability assays, and PEGASUS[31] provides the contrasting case in which selected designs were synthesized and assayed for solubility and MDCK-MDR1 permeability. The strongest claim remains the one for which the generated chemical space, optimized objective, and measured assay are aligned.

Predictor applicability is therefore the controlling uncertainty across this category. CycPeptMPDB supplies a large curated source of cyclic-peptide permeability records[98], but source and assay imbalance remain important. CycPeptMP[99], Multi_CycGT[100], MuCoCP[101], the representation study of Cabezon and co-workers[102], MultiCycPermea[103], and PCPpred[104] illustrate different sequence, graph, multimodal, and multi-assay prediction strategies. A descriptor-based macrocycle model provides a further transferable scoring baseline[105]. None should be treated as universally calibrated across new scaffolds, topologies, stereochemistries, and assay formats. Monte Carlo dropout offers a lightweight Bayesian approximation[106], deep ensembles provide a scalable alternative[107], and conformal prediction supplies coverage-oriented uncertainty estimates under exchangeability assumptions[108].

Developability beyond permeability is less mature for the models discussed here. PlifePred models blood half-life for natural and modified peptides[25], a simple physicochemical regression estimates serum half-life for unmodified sequences[109], and transfer learning with enzymatic-cleavage features extends half-life prediction across species and organs[26]. HAPPENN predicts hemolytic propensity[27], while Hemolytik provides experimentally annotated hemolytic and non-hemolytic sequences[110]; these safety resources are dominated by linear or canonical peptides and therefore offer methodological transfer rather than validated constrained-peptide rewards. For multi-objective strategy, MOFormer[111] uses Pareto-based feedback, whereas HMAMP[112] uses a hypervolume-driven objective. Both show how trade-offs can be preserved without collapsing activity, hemolysis, and toxicity into one weighted score, while constrained topology and macrocycle developability remain outside their demonstrated scope.

Pareto search, direct preference optimization, reinforcement-learning rewards, and Bayesian or conformal acquisition become more persuasive when chemical validity is enforced, uncertainty and applicability are incorporated, and the optimized endpoint is tested prospectively. Within the surveyed literature, few constrained-peptide programs combine these elements in a single evidentiary frame. The strongest studies therefore report generation, predictor validation, domain analysis, and assay measurement together.

Cross-model comparison and methodological limitations

Across model families, representation creates a persistent capability trade-off rather than a simple hierarchy of model quality. HELM-GPT[39] preserves curated monomer semantics and explicit bonds, PepINVENT[40] exposes more submonomer chemistry, and SELFIES[41] demonstrates syntax robustness for general molecular strings. Residue-level structure models encode target and closure priors while compressing atomistic chemistry. CpSDE[76] broadens explicit connectivity at atom level, whereas NCFlow[67] focuses on noncanonical side-chain placement. Within the surveyed literature, few representations preserve residue semantics, broad covalent chemistry, stereochemical fidelity, and efficient multi-state geometry simultaneously.

Topology is represented with similarly uneven fidelity. Head-to-tail adjacency can be encoded compactly, whereas side-chain linkages, mixed crosslinks, and multicyclic scaffolds require compatible chemistry, closure geometry, stereochemistry, and local strain to agree. GeoCycler shows that multicyclic constraint satisfaction remains more difficult than simple closure[74]. Stereogenic all-atom prediction can improve chemical fidelity[72], but a valid graph or satisfied distance restraint does not by itself establish a strain-free or synthetically accessible macrocycle. Force-field benchmarks show why local energetic plausibility must be evaluated rather than assumed[37].

Structural realism also depends on the object a model was built to generate. Target-conditioned systems often emphasize one bound pose, conformer generators sample unbound or simulation-like states, and property optimizers commonly operate on two-dimensional descriptions. For chameleonic peptides, aqueous, membrane-competent, and target-bound populations may differ[23]. Membrane-transit studies emphasize solvent-dependent redistribution[22], while conformational-sampling work shows that relevant macrocycle states can depend strongly on environment[87] and molecular flexibility[113]. Pipelines that combine these models are most informative when they preserve molecular identity and uncertainty across state changes rather than treating the outputs as interchangeable structures.

Evidence levels remain family-specific and should be compared as claims, not as interchangeable scores. RFpeptides reports solved target complexes and biochemical binding[57], AfCycDesign reports solved designed structures and binding assays[56], and CYC_BUILDER adds target-specific cellular functional assays[60]. PEGASUS reports prospective synthesis, solubility measurements, and MDCK-MDR1 permeability for selected designs, but with limited target context[31]. Most other claims in the surveyed literature rely on formal validity, closure, docking, energy, retrospective property prediction, or short molecular-dynamics trajectories. Designed membrane-traversing macrocycles provide a rare coherent program in which structure, permeability, and in vivo exposure were assessed together[9]. Scores or assays from separate studies should not be combined to imply that one molecule satisfies binding, permeability, stability, pharmacokinetics, and synthesis requirements. Even in experimentally tested studies, candidate numbers, scaffold breadth, prospective replication, and synthetic-feasibility reporting are often limited.

Rule embedding also varies in strength and location. Representation-level topology, constrained decoding, denoising guidance, preference objectives, and posterior guidance can directly shape the sampled distribution; relaxation, ranking filters, docking, and external predictors usually evaluate or reorder candidates after proposal. Rosetta-based

Table 3 | Multidimensional evaluation and reporting framework for constrained-peptide generative systems

Dimension	Operational question	Minimum reporting	Interpretive boundary
Chemical identity, validity and stereochemical fidelity	Fully specified and valid before repair?	Valid fractions before and after repair; valence, formal charge, residue-to-atom mapping and exact stereochemistry	Parsing or repair does not establish synthesizability or three-dimensional plausibility.
Topology compliance	Intended covalent topology satisfied?	Closure rates by topology; correct linkage atoms, bond order and ring count; failures by ring size; closure geometry or strain	Distance compliance alone does not establish chemically valid, low-strain closure.
Representation fidelity	Explicit, inferred, reconstructed or absent?	Round-trip molecular identity; isomer, stereochemistry and topology preservation; losses at model handoffs	Do not attribute downstream-reconstructed information to the generator.
Three-dimensional plausibility	Physically credible, task-compatible coordinates?	Clashes, strain and bond geometry; independent structural comparisons; multiple seeds; confidence calibration	Confidence, docking scores or brief relaxation alone do not validate structures.
Ensemble and environment fidelity	Relevant state populations reproduced across conditions?	Conformer coverage and precision; distributional metrics; population uncertainty; environment labels; available independent observables	Diversity or low energy does not establish Boltzmann or environment-dependent population fidelity.
Target-specific performance	Intended target engagement and defined mechanism?	Target holdout; separate docking, biochemical and cellular results; affinity or function; negatives and replicates	Predicted poses do not establish binding; cellular responses can conflate binding, uptake, stability and downstream biology.
Developability	Assay-aligned endpoint and in-domain prediction?	Endpoint and assay identity; split strategy; calibration and applicability domain; separate predicted and measured results	Score gains are not measured developability; assays are not interchangeable.
Generalization and uncertainty	Claims supported across constrained-peptide classes?	Results and sample sizes by topology, ring size, linkage, stereochemistry and noncanonical content; OOD holdouts; coverage and abstention	Aggregate metrics can hide class-specific failure; uncertainty guarantees remain assumption-dependent.
Synthetic tractability	Candidates synthesizable and analytically confirmed?	Monomer and reaction constraints; attempted and successful syntheses; yield, purity, identity and failure reasons	Generic accessibility scores miss peptide coupling, macrocyclization, purification and monomer-supply constraints.
Evidence level and pipeline attribution	Endpoint-specific evidence and contributing pipeline components?	Evidence level for each claimed endpoint; complete candidate funnel; fixed budgets; prespecified criteria; matched ablations; generated, synthesized and tested counts; failures and success rates	Evidence is endpoint-specific; hits validate the composite workflow unless generator, selection and experimental contributions are separated.

Dimensions are reported as a standardized evidence-and-reporting profile rather than collapsed into a single score. "Not evaluated" should be used when a method does not claim or test a dimension. All rates require explicit denominators, and subgroup results should include sample sizes and uncertainty. Across applicable dimensions, reports should identify the data and assay source, preprocessing and split definition, metric implementation, numbers of generated samples and independent runs, random seeds, sampling or inference budget, model version or checkpoint, and uncertainty estimates. For each claimed endpoint, specify the supporting evidence, distinguishing computational predictions from biochemical, cellular or in vivo measurements where applicable.

relaxation and design are powerful but external unless coupled into the generative objective[34]. Docking remains program-dependent[35], and generic generative benchmarks such as GuacaMol[43] do not test peptide closure or stereochemistry. Model reports should therefore state where each rule enters and which failure modes are left to evaluation.

Surrogate optimization should be interpreted through validation design rather than score gain alone. Sparse and heterogeneous cyclic-peptide assays make applicability domains narrow[20]; activity cliffs can turn small edits into misleading score jumps[28]; prospective ADME studies show that transfer can fail[29]; and real-world molecular shifts may not resemble standard benchmark out-of-distribution settings[114]. Leakage-resistant splits, scaffold or topology holdouts, uncertainty intervals, explicit out-of-domain analysis, and prospective synthesis or assay are therefore not optional additions. They define the level of claim that an optimized score can support.

Out-of-distribution failure depends on the represented object. String models may remain syntactically valid while combining rare modifications or linkages in unsupported chemical contexts[114]. Residue-level geometry can satisfy a coarse closure restraint yet imply atomistic clashes, strained bonds or incompatible stereochemistry; atom-bond generators may preserve local valence while missing global ring strain or accessible closure[77]. Conformer models can collapse towards familiar ring sizes, omit low-population states or reproduce biases in simulation-derived training ensembles[37]. Property-guided search has a different pathology: optimization can populate regions favored by a surrogate even as calibration and assay relevance deteriorate.

Applicability analysis should therefore be conditional on topology, ring size, linkage chemistry, stereochemical composition, modification burden and assay format. Pre-specified family holdouts, distances to training data, calibration curves, ensemble disagreement and abstention rates should be reported globally and within these strata. Conformal intervals can make refusal or qualification auditable, but marginal coverage under exchangeability does not guarantee reliability under unrestricted chemical shift[20]. Unsupported classes are more accurately described as outside the demonstrated scope than absorbed into an aggregate performance estimate.

Multidimensional evaluation and pipeline attribution

Objective comparison requires a capability profile rather than a scalar ranking. Chemical identity, topology, stereochemistry, three-dimensional geometry, ensemble fidelity, target engagement, developability, synthetic tractability, uncertainty and experimental evidence answer different questions. A method should be assessed on the object and endpoints it claims to generate; an untested dimension is "not evaluated", not an inferred success or a zero. Comparisons within a dimension require commensurate objects, endpoints, data provenance and splits, denominators and evaluation budgets; otherwise, evidence profiles should be presented separately.

Table 3 converts these dimensions into minimum reporting requirements. Object-level metrics should precede property or experimental endpoints, and every rate should retain its denominator: validity before repair, closure by topology, stereochemical error, conformer coverage against an identified reference, and assay success among compounds actually synthesized and tested. Subgroup results require sample sizes and uncertainty, especially for small or chemically unusual classes. Reproducibility also requires disclosure of the data and assay source, preprocessing and split definition, metric implementation, numbers of generated samples and

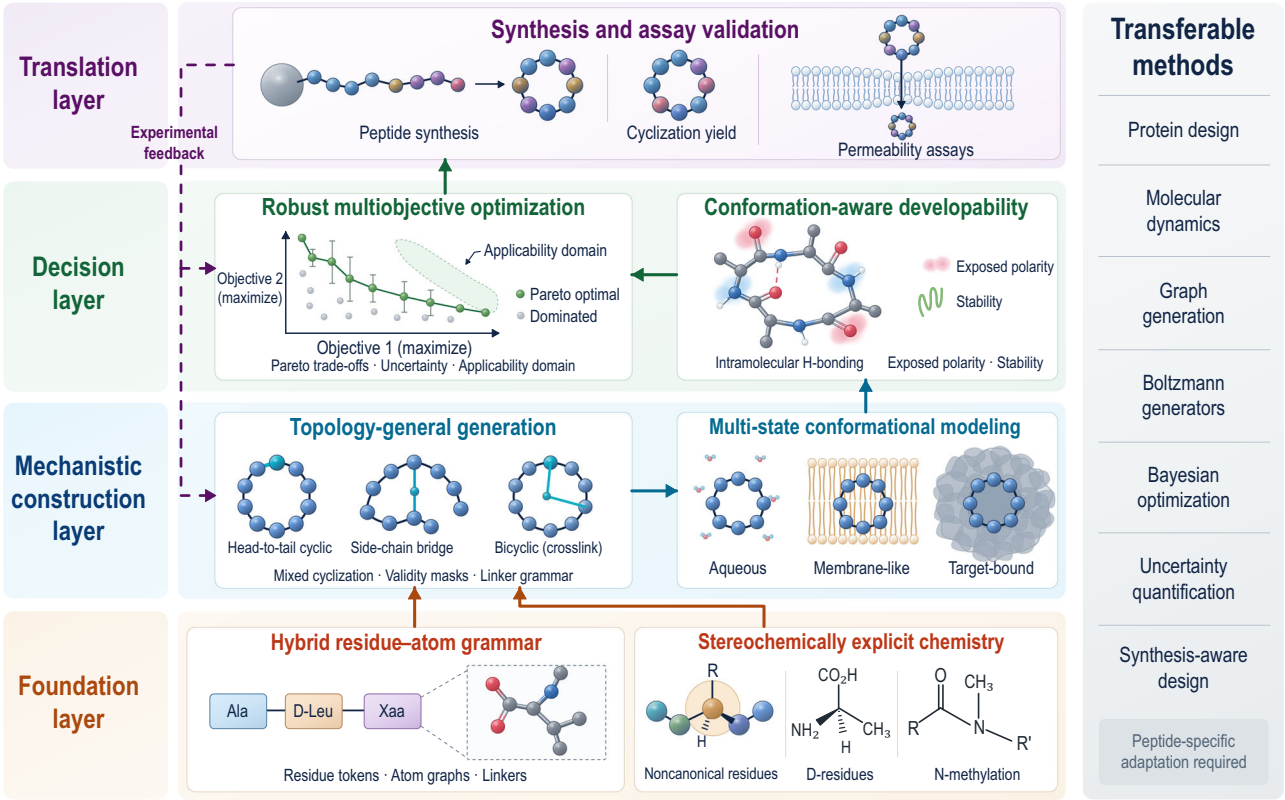


Fig. 4 | Proposed dependencies and experimental feedback in constrained-peptide design. Chemical representations support topology generation and multi-state conformational modeling, which inform developability assessment, multi-objective optimization and experimental validation. Solid arrows indicate proposed dependencies; dashed purple arrows return experimental feedback to optimization and topology generation. The sidebar lists transferable methods requiring peptide-specific adaptation. In the Pareto plot, green and gray points denote nondominated and dominated candidates; error bars illustrate uncertainty, and the shaded region represents an applicability domain. Molecular depictions, membrane transport and optimization plots are schematic, not quantitative data. In the atomistic drawings in the hybrid residue–atom grammar and conformation-aware developability modules, gray, blue, red and white spheres denote carbon, nitrogen, oxygen and hydrogen, respectively. In the residue tokens, Ala denotes alanine, D-Leu denotes D-leucine, and Xaa denotes an unspecified residue. R and R′ denote variable substituents.

independent runs, random seeds, sampling or inference budget, model version or checkpoint, and uncertainty estimates.

End-to-end success must also be separated from generator performance. A peptide emerging from extensive sampling, predictor ranking, structural refinement, expert inspection and experimental screening validates the composite workflow, but does not identify the responsible component. Reports should retain the candidate funnel from raw proposals to tested compounds, including counts entering and leaving each stage, failure reasons, manual interventions and any reuse of experimental information during selection.

Where feasible, matched-budget ablations provide a direct route to attribution. Generator-only output, generator plus property scoring, generator plus structural refinement and the complete workflow can be compared under the same sampling and testing budget. When full prospective ablation is impractical, reports should at minimum provide the complete candidate funnel and computational ablations under a fixed sampling budget. Selection rules and acceptance thresholds should be fixed before outcomes are known; blinded or externally held-out experiments provide a stronger test than post hoc enrichment of a favorable subset.

Overall, each family contributes useful components, but cross-model comparison exposes how often molecular identity, uncertainty, and evidence level are weakened at handoffs between representation, geometry, property prediction, and validation. The next section therefore turns from diagnosis to a staged dependency roadmap for more coherent constrained-peptide design systems.

Residual bottlenecks and model evolution roadmap

The remaining bottlenecks form a dependency chain. Chemical identity and stereochemistry determine what can be built; topology and geometry determine what can close and bind; conformational populations determine which physicochemical properties are expressed; decision models negotiate competing objectives; and synthesis-assay feedback tests whether the computational hypothesis survives translation. Figure 4 summarizes these proposed dependencies and the experimental feedback linking later stages to earlier design decisions. Improvements at later layers are unlikely to compensate for ambiguous chemistry or invalid topology inherited upstream.

A next-generation representation would benefit from enforcing molecular identity wherever ambiguity is unacceptable. HELM-GPT[39] and PepINVENT[40] show complementary residue-level grammars; CpSDE[76] provides atom-bond topology; NCFlow[67] and UNAAGI[68] illustrate local atom-level handling of noncanonical side chains. A hybrid residue-atom grammar could retain peptide-scale semantics and efficient sequence planning while using local atom graphs to resolve N-methylation, D- and β-amino acids, thioethers, lactams, staples, multicyclic linkers, and stereochemistry. The relevant benchmark is faithful decoding of identity and connectivity, not novelty alone.

Once chemical identity is stable, generated geometry should remain compatible with both covalent topology and conformational multiplicity. RINGER[77] provides internal-coordinate ensemble generation, the Bayaraa model[83] learns mixed-chirality dihedral distributions, CPGEN[78] conditions on ensemble-level backbone statistics, and CyclicCAE[84] supplies a latent

backbone representation. These methods are better viewed as building blocks for multi-state design than as complete therapeutic-design systems. Closure should emerge with bond-compatible coordinates, and aqueous, membrane-like, and target-bound states should be represented when mechanistically relevant. A single plausible structure would not, by itself, support an ensemble-level claim; topology should remain intact across sampled states and population estimates should have a physically interpretable reference.

Property optimization becomes more defensible when it consumes state-dependent features and reports the limits of the estimator. PEGASUS combines assay-derived and quantum-mechanical descriptors[31], applicability-domain analysis identifies where permeability predictions are supported[20], and systematic benchmarking exposes variation among current AI methods[24]. Chameleonicity metrics provide a framework for quantifying environment-dependent polarity[115], while descriptor-based macrocycle models illustrate the value and limits of simpler scoring functions[105]. Rather than returning one maximized score, a decision layer could provide calibrated trade-offs, uncertainty estimates, and the scaffold, topology, assay, and chemical domain in which each estimate is supported.

The loop closes through peptide-specific synthesis and assays. Bulk live-cell permeability measurement can generate experimentally grounded labels at scale[32]. Self-driving-laboratory principles provide a general template for automated experiment selection[116], and self-play reinforcement learning in protein engineering illustrates how iterative experimental feedback can alter design policies[117]. Generic synthetic-complexity scores such as SCScore[118] are useful comparators but do not capture monomer availability, coupling difficulty, macrocyclization outcome, purification burden, or analytical confirmation in peptide synthesis. Small batches may be especially informative when chosen to distinguish competing model hypotheses, test an out-of-distribution region, or calibrate uncertainty rather than simply confirm the top-ranked score.

This ordering also clarifies failure diagnosis. Chemical or stereochemical errors should remain distinguishable from topology failure; ensemble mismatch should not be relabeled as a property-prediction error; and unsuccessful synthesis should not be used to infer that a binding model was wrong. Layer-specific attribution is important if experimental feedback is to update the component that actually failed.

Interoperability turns these layers into a design system. Ideally, a monomer or bond edit would propagate without loss of identity into coordinates, ensemble populations, property features, and synthetic annotations. CPSea provides target-complex training data[80], CREMP provides macrocyclic conformer-rotamer ensembles[86], SPICE supplies diverse atomistic molecular and peptide configurations for learned potentials[119], and CyclicPepedia organizes natural and synthetic cyclic-peptide knowledge[120]. Because these resources differ in provenance and domain, transferred priors should carry explicit data-source and applicability labels.

Staged integration is likely to be more tractable than an all-purpose monolith. Near-term systems could pair a chemical grammar with hard validity constraints, one geometric regime, and one well-defined assay domain, then add environment-conditioned ensembles and broader objectives as lower-level performance stabilizes. Useful success criteria include retained rule compliance across interfaces and prospective evidence, rather than the number of modules assembled.

Future directions

The most tractable advances lie at identity-preserving interfaces where one capability hands off to another. Progress should therefore be judged not only by adding modules, but by whether chemical identity, stereochemistry, topology, conformational state, assay context, and uncertainty remain traceable across the design workflow. Useful next steps are those that broaden topology, connect conformational states to measured properties, and use experiments to challenge rather than merely confirm model rankings.

Topology selection should move inside stereochemically explicit chemical grammars rather than remain a short menu of predefined closures. Existing stereogenic, reward-aligned, automated-linkage, and atom-bond generative approaches provide partial ingredients[72,74–76], but the next step is to let target and property context choose among head-to-tail, side-chain, disulfide, thioether, stapled, and multicyclic solutions under shared validity rules. Residue-level planning should therefore be coupled to atom-level valence masks, chiral constraints, and reaction-compatible linkage templates. Conservative validity gates would be essential, because a larger topology space is useful only if generated candidates remain chemically interpretable.

Environment-conditioned ensembles provide a direct bridge between geometry and developability. Current internal-coordinate, flow-matching, mixed-chirality diffusion, and autoencoder methods offer useful ensemble primitives[77,78,83,84], but water, membrane-like media, and target-bound states are not yet commonly linked within one objective. Future models could preserve a shared molecular identity across condition-specific ensemble heads, with population losses anchored by simulation, NMR, hydrogen-exchange, crystallographic, or permeability-related observables. Because conformational sampling can miss relevant states in flexible macrocycles[113], agreement among several independent physical signals would be more convincing than reliance on one energy model.

Environment-conditioned ensembles also impose a throughput constraint that is absent from single-state objectives. Cost scales with candidate count, environmental states, conformers per state and the physical evaluations applied to each conformer; repeated relaxation or enhanced sampling is therefore difficult to reconcile with unrestricted library-scale search[83]. Learned conformer generators and conditioned Boltzmann models[82] can amortize part of this cost, but they inherit assumptions from the reference sampler, force field and training domain[37].

A tiered design loop can allocate fidelity selectively. Two-dimensional or language-model predictors provide inexpensive triage, applicability and uncertainty gates determine when those estimates are credible, learned ensemble surrogates resolve the smaller set for which state dependence matters, and molecular simulation, quantum calculations or experiment are reserved for calibration and final discrimination. Wall-clock or accelerator cost, the number of generated states and conformers, and the fraction advanced at each tier are necessary to judge operational scalability.

Developability models should increasingly distinguish static descriptors from state-dependent mechanisms. Physics-based permeability work shows that transfer free energy and polar shielding can outperform simple intramolecular-hydrogen-bond counts[121], while PEGASUS illustrates the value of assay-derived and multisolvent quantum-mechanical features[31]. Quantitative chameleonicity metrics frame environment-dependent exposed polarity[115], and bulk live-cell assays provide an experimental endpoint for calibration[32]. Future ensemble encoders could combine exposed polar surface area, hydrogen-bond occupancy, conformer populations, assay metadata, and calibrated uncertainty. Outputs should remain assay-specific, because permeability, oral absorption, proteolysis, and cell entry reflect different mechanisms.

Multi-objective search should preserve trade-offs without preferentially rewarding unsupported extrapolation. Pareto maintenance and Pareto-based feedback[92,111], preference optimization[95], hypervolume feedback[112], and conformal scoring provide transferable strategies[90], but they become more persuasive only when coupled to chemical validity, topology validity, scaffold-aware uncertainty, and applicability constraints. Candidate ranking should therefore consider both predicted utility and evidence quality, allowing a well-supported moderate proposal to outrank an extreme out-of-domain candidate while reserving part of the acquisition budget for informative exploration.

Physics-based guidance is most useful as a calibrated prior rather than an oracle. A comparative study of ten docking programs documents substantial method dependence for protein-cyclic-peptide complexes[35], and the HADDOCK2.4 cyclic-peptide protocol shows that docking success depends on peptide ensemble quality and receptor state[122]. Rosetta supplies mature energy and design machinery[34], but standard force fields such as CHARMM36m[81] were not parameterized specifically for all macrocycle

chemistries, and direct cyclic-peptide force-field benchmarks reveal population errors[37]. Enhanced sampling can improve retrospective permeability correlations[89], yet combining several partially independent physical signals with learned uncertainty may be less brittle than optimizing a single energy function.

Synthesis-aware generation likewise requires peptide-specific process information rather than generic small-molecule accessibility scores. SCScore estimates synthetic complexity from reaction-corpus precedents[118], but it does not encode solid-phase coupling efficiency, protecting-group orthogonality, aggregation, macrocyclization yield, purification, or monomer supply. Reaction-compatible templates can exclude impossible bonds, while probabilistic yield and purification models retain room for novel but uncertain chemistry. Historical bias should be monitored so that a model trained on routine synthesis does not automatically exclude useful innovation.

Prospective experiments are most informative when they reduce uncertainty rather than simply identify active compounds. Bulk live-cell permeability measurements[32], self-driving-laboratory frameworks[116], and self-play protein-engineering workflows provide useful precedents for calibration[117], experiment selection, and iterative policy updating. In constrained-peptide design, acquisition should balance predicted utility with scaffold diversity, mechanistic discrimination, replicate-aware assay metadata, and uncertainty reduction. This would keep experimental feedback tied to the model component it is meant to update rather than collapsing results into a context-free label.

Benchmarking should distinguish object validity, mechanistic fidelity, and translational evidence. GuacaMol provides a general framework for molecular-generation benchmarks[43], but constrained peptides additionally require scaffold-, topology-, stereochemistry-, and modification-aware splits. Applicability-domain analysis[20] and systematic permeability benchmarking[24] should be paired with target-complex holdouts from resources such as CPSea[80] and ensemble holdouts from CREMP[86]. Reports should state which rules shape generation, which are applied afterward, and which endpoints are experimentally measured. Retrospective performance remains useful, but blinded synthesis and assay tests provide a stronger assessment of generalization.

Decision support should expose the rule path for each candidate: chosen topology, chemical edits, dominant conformational states, property trade-offs, uncertainty, and synthetic concerns. PepExplainer provides an example of explainable macrocyclic-peptide bioactivity prediction and selection-based optimization[123]. Such a record can help medicinal chemists distinguish designs supported by chemistry and data from those driven mainly by a fragile predictor, while making expert overrides auditable. Interpretability is most informative when it reflects variables used by the model rather than a post hoc narrative disconnected from the computation.

Outlook

Constrained-peptide generation is moving from valid-object construction toward coordinated models of chemical grammar, target geometry, covalent topology, conformational state, and therapeutic objective. The useful question is not which generic architecture is used, but where a physicochemical rule changes the distribution: in representation, conditioning, denoising, flow, decoding, preference optimization, or experimental feedback.

The five model families are complementary rather than interchangeable. Sequence, notation, and latent-space methods provide scalable chemical language; target-conditioned design contributes structural and binding hypotheses; geometric and atom-bond models encode or guide closure and connectivity; ensemble methods represent state populations; and developability-guided systems negotiate predicted or measured property trade-offs. Convergence will be most meaningful when each component carries its validity conditions, evidence level, and uncertainty into the next.

Translation calls for a stricter standard than computational plausibility. A generated constrained peptide becomes more therapeutically informative when its chemistry is unambiguous, its relevant states support the proposed mechanism, its property estimates are calibrated within domain, and synthesis and assays test the same claim. Selective prospective validation is therefore part of the design loop rather than an interchangeable final score.

The central opportunity is to generate fewer candidates with stronger internal coherence. Topology, chemistry, conformational ensemble, target engagement, developability, and synthetic route are most useful when they form one testable hypothesis. Physicochemical rule embedding offers both a framework for building such systems and an audit trail for judging the claims they produce.

Data availability

No new research datasets were generated in this Review. Publicly available structural coordinates of the cyclophilin A–cyclosporin A complex (PDB ID: 1CWA; https://www.rcsb.org/structure/1CWA) were used solely to prepare the structural illustrations in Fig. 3c, not to generate new research findings.

Code availability

Code availability is not applicable to this Review because no custom code was generated or used to support its main claims.

Received: 12 July 2026; Accepted: 29 September 2026;
Published online: 29 October 2026

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Acknowledgements

Not applicable.

Author contributions

Y.W. conceived the review, curated the literature, prepared the figures and tables, and drafted the manuscript. D.C. provided critical input on the pharmaceutical and developability aspects of constrained peptide therapeutics. X.Z. contributed to the interpretation of generative modeling methods. X.F. and X.Y. supervised the work, guided the conceptual framework, and revised the manuscript. All authors reviewed and approved the final version.

Funding

This work was supported in part by the National Science and Technology Major Project for Innovative Drug Research and Development (No. 2025ZD1802302 to D.C.) and the National Natural Science Foundation of China (Nos. 62572178 and 62372158 to X.F.). The funder had no role in the selection of the review topic, literature review, manuscript preparation, or the decision to submit the manuscript for publication.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Xiangzheng Fu or Xiaojun Yao.

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